

Dhyan Suresh

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EDUCATION

University of Central Florida
Bachelor of Science in Computer Science

Expected Dec 2027
Orlando, FL

Santa Fe College
Associate in Arts

May 2024
Gainesville, FL

EXPERIENCE

Undergraduate Researcher
Knight Research Scholars Program, University of Central Florida

Sep 2026 – Present
Orlando, FL

- Built a Python ankle-angle estimator with **NumPy** to produce estimates **within 5°** of ground-truth angles
- Reduced initial ankle-angle estimation error (RMSE) by **98.3%** by estimating foot and shank IMU bending axes from three-axis gyroscope data and filtering drift
- Processed **392,360 IMU samples** at 2,000 Hz and aligned them with 100 Hz OpenSim data using **SciPy Butterworth filtering**

Marketing Lead
Google Developer Student Club at UCF

Aug 2024 – Apr 2025
Orlando, FL

- Grew the club's social media following by **50%** through consistent event promotion and outreach
- Promoted **12+** technical workshops over two semesters, averaging **20** student attendees per event
- Coordinated weekly event planning and promotions for a **five-member team**, tracking deliverables in Trello

Network Technician
West Networks

May 2023 – May 2024
Gainesville, FL

- Built and deployed **100+** portable Peplink BR2 Pro/BR1 routers with dual lithium batteries, providing reliable remote connectivity for field sites
- Configured **dual-SIM** protocol enabling auto-switching between cell carriers during link failures
- Troubleshoot field-site technical issues using Peplink InControl to diagnose cellular link failures on deployed routers

PROJECTS

SignSpeak | *React, PyTorch, MediaPipe, Python*

- Developing an American Sign Language interpreter to recognize letters and common phrases in real time via webcam, with spoken output through the **Web Speech API**
- Engineered a client-side ML pipeline via **MediaPipe Hands** to extract **21** hand landmarks in 3D per frame
- Building a **Python** processing pipeline to normalize landmark coordinates from 87K+ image dataset (ASL Alphabet) and train a multi-layer classifier in **PyTorch**

SafeFlight | *React, Node.js, PostgreSQL, Leaflet* | safeflight.onrender.com

- Built a **React + TypeScript** client plotting live aircraft positions on a Leaflet map, integrating 3 **REST APIs** (AeroDataBox, OpenSky, aviationweather.gov)
- Engineered a **Node.js/Express** API with Google OAuth sign-in, restricting flight data to each user and accepted friends
- Designed **PostgreSQL** database via **Prisma** to store users, flights, and friends list

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C, HTML/CSS

Frameworks & Libraries: React, Node.js, Express, FastAPI, NumPy, PyTorch, MediaPipe, SciPy

Databases: PostgreSQL, MongoDB, Firebase Firestore

Developer Tools: Git, GitHub, Vercel, Docker, Figma, Render